Module 5 Assignment

SQL Functions

Database Setup

Employees Table

```sql
CREATE TABLE Employees
(
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    Salary DECIMAL(10,2),
    DeptID INT,
    HireDate DATE,
    Bonus DECIMAL(10,2) NULL
);

INSERT INTO Employees VALUES
(101,'Rahul Sharma',35250,1,'2021-05-10',2000),
(102,'Priya Mehta',55800,2,'2020-08-15',NULL),
(103,'Amit Verma',70500,3,'2019-03-22',5000),
(104,'Sneha Reddy',44800,1,'2022-01-12',NULL),
(105,'Karan Patel',81200,3,'2018-07-05',7000),
(106,'Neha Singh',29900,4,'2023-02-18',1500),
(107,'Arjun Rao',61000,2,'2021-11-03',NULL),
(108,'Pooja Nair',39500,4,'2022-07-21',1000);
```

Assignment Questions

Question 1

The HR team wants a formatted employee directory for internal use.

Create a query that displays:

- Employee ID
- Employee Name in **UPPERCASE**
- First 4 characters of the employee name
- Salary converted to **integer**
- Hire Date in **dd-MM-yyyy** format

Expected Output Columns

```None
EmpID | EmployeeName | NamePrefix | SalaryAsInteger |
FormattedHireDate
```

Question 2

The Finance team wants a salary review report.

Create a query to display:

- Employee Name
- Original Salary
- Salary rounded to nearest 1000
- Difference of salary from ₹50,000 using an absolute value
- Salary Band using the below rules:

```None
Salary >= 70000         → Senior Band
Salary between 40000 and 69999 → Mid Band
Salary below 40000      → Entry Band
```

Expected Output Columns

```None
EmpName | Salary | RoundedSalary | SalaryDifference | SalaryBand
```

Question 3

Management wants to track employee tenure and joining details.

Create a query that displays:

- Employee Name
- Hire Date
- Number of months completed in the company
- Joining month name
- End date of the joining month

Expected Output Columns

```None
EmpName | HireDate | MonthsCompleted | JoiningMonth | MonthEndDate
```

Question 4

The payroll team wants to prepare a clean bonus report.

Create a query that displays:

- Employee Name
- Original Bonus
- Bonus after replacing NULL with 0
- Bonus Status:
    - `No Bonus` if bonus is NULL
    - `Bonus Available` otherwise

Expected Output Columns

```None
EmpName | Bonus | FinalBonus | BonusStatus
```

Question 5

The HR team wants to identify the top 2 highest-paid employees in each department.

Write a query that displays:

- **Employee Name**
- **Department ID**
- **Salary**
- **Rank of employee salary within each department**

Requirement

- **Use a window function**
- **Rank employees department-wise**
- **Return only the top 2 employees from each department**

Expected Output Columns

```None
EmpName | DeptID | Salary | DeptRank
```

Question 6

The HR team wants to identify the highest paid employees **within each department**.

Create a query that displays:

- Employee Name
- Department ID
- Salary
- Rank of employee salary within each department

Requirement

Use a window function with **PARTITION BY DeptID**.

Expected Output Columns

```None
EmpName | DeptID | Salary | DeptSalaryRank
```

Question 7

Management wants to compare each employee's salary with the salary of the employee listed just before them when sorted by salary.

Create a query that displays:

- Employee Name
- Salary
- Previous Employee Salary
- Difference between current salary and previous salary

Requirement

Use **LAG()**.

Expected Output Columns

```None
EmpName | Salary | PreviousSalary | SalaryDifference
```

Question 8

The finance team wants a department-wise salary summary along with the company grand total.

Write a query to display:

- Department ID
- Total Salary

The result should include:

- individual department totals
- one grand total row

Requirement

Use **ROLLUP**.

Expected Output Format

```None
DeptID | TotalSalary
```